H2: Determinant (uitgewerkte oplossingen)

1) oef 10 (determinant van Vandermonde)

```
(%i1) V: matrix(
        [1,a,a^2],
        [1,b,b^2],
        [1,c,c^2]
      );
```

(V) $\begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$

```
(%i2) determinant(V);
```

(%o2) $-a\left(c^2-b^2\right)+b\,c^2+a^2\left(c-b\right)-b^2\,c$

```
(%i3) factor(%);
```

(%o3) $\left(b-a\right)\left(c-a\right)\left(c-b\right)$

2) oef 11

2.1 (a)

```
(%i4)  A: matrix(
         [8,1,9,1],
         [6,3,4,5],
         [5,4,7,2],
         [2,3,4,1]
       );
```

(A) $$\begin{bmatrix} 8 & 1 & 9 & 1 \\ 6 & 3 & 4 & 5 \\ 5 & 4 & 7 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix}$$

```
(%i5)  determinant(A);
(%o5)  20
```

2.2 (b)

```
(%i6)  B: matrix(
         [1,5,2,6,9],
         [2,4,2,4,8],
         [3,3,2,1,7],
         [4,2,2,7,2],
         [5,1,1,9,9]
       );
```

(B) $$\begin{bmatrix} 1 & 5 & 2 & 6 & 9 \\ 2 & 4 & 2 & 4 & 8 \\ 3 & 3 & 2 & 1 & 7 \\ 4 & 2 & 2 & 7 & 2 \\ 5 & 1 & 1 & 9 & 9 \end{bmatrix}$$

```
(%i7)  determinant(B);
(%o7)  24
```

2.3 (c)

```
(%i8)  C: matrix(
         [4,7,1,5],
         [3,-1,2,1],
         [2,0,3,6],
         [1,-4,4,-3]
       );
```

(C) $$\begin{bmatrix} 4 & 7 & 1 & 5 \\ 3 & -1 & 2 & 1 \\ 2 & 0 & 3 & 6 \\ 1 & -4 & 4 & -3 \end{bmatrix}$$

```
(%i9)  determinant(C);
(%o9)  405
```

2.4 (d)

```
(%i10)  D: matrix(
          [3,5,6,-2,4],
          [-4,0,2,3,0],
          [2,-3,0,1,2],
          [4,5,4,-2,7],
          [-2,6,0,0,3]
        );
```

(D) $$\begin{bmatrix} 3 & 5 & 6 & -2 & 4 \\ -4 & 0 & 2 & 3 & 0 \\ 2 & -3 & 0 & 1 & 2 \\ 4 & 5 & 4 & -2 & 7 \\ -2 & 6 & 0 & 0 & 3 \end{bmatrix}$$

```
(%i11)  determinant(D);
(%o11)  -390
```

3) oef 12

3.1 (a)

(%i12)
```
A: matrix(
 [3*x-y,2*x,2*x],
 [2*y,x+y,2*y],
 [4*x,4*x,3*x+y]
);
```

(A) $$\begin{bmatrix} 3x-y & 2x & 2x \\ 2y & y+x & 2y \\ 4x & 4x & y+3x \end{bmatrix}$$

(%i13)
```
determinant(A);
```

(%o13) $$(3x-y)\left((y+x)(y+3x)-8xy\right)-2x\left(2y(y+3x)-8xy\right)+2x\left(8xy-4x(y+x)\right)$$

(%i14)
```
factor(determinant(A));
```

(%o14) $$-(y-x)^3$$

3.2 (b)

(%i15)
```
B: matrix(
 [p,2,p+1],
 [3,4,5],
 [2,p,1]
);
```

(B) $$\begin{bmatrix} p & 2 & p+1 \\ 3 & 4 & 5 \\ 2 & p & 1 \end{bmatrix}$$

(%i16)
```
determinant(B);
```

(%o16) $$(p+1)(3p-8)+(4-5p)p+14$$

(%i17)
```
factor(determinant(B));
```

(%o17) $$-(p+2)(2p-3)$$

3.3 (c)

(%i18)
```
C: matrix(
 [a,a+1,a+2],
 [a+3,a+4,a+5],
 [a+6,a+7,a+8]
);
```

(C) $$\begin{bmatrix} a & a+1 & a+2 \\ a+3 & a+4 & a+5 \\ a+6 & a+7 & a+8 \end{bmatrix}$$

Tip: eerst kijken, dan rekenen (R2-R1=R3-R2, dus det(C)=0)

Ter controle:

(%i19)
```
determinant(C);
```

(%o19) $$a\left(\left(a+4\right)\left(a+8\right)-\left(a+5\right)\left(a+7\right)\right)-\left(a+1\right)\left(\left(a+3\right)\left(a+8\right)-\left(a+5\right)\left(a+6\right)\right)+\left(a+2\right)\left(\left(a+3\right)\left(a+7\right)-\left(a+4\right)\left(a+6\right)\right)$$

(%i20)
```
factor(%);
```

(%o20) 0

3.4 (d)

(%i21)
```
D: matrix(
 [a,b,c],
 [b,c,a],
 [c,a,b]
);
```

(D) $$\begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$$

```
(%i23) determinant(D);
       factor(%);
```

(%o22) $c\left(a\,b-c^2\right)+a\left(b\,c-a^2\right)-b\left(b^2-a\,c\right)$

(%o23) $-\left(c+b+a\right)\left(c^2-b\,c-a\,c+b^2-a\,b+a^2\right)$

3.5 (e)

```
(%i24) E: matrix(
         [a,b,b,b],
         [b,a,b,b],
         [b,b,a,b],
         [b,b,a,b]
       );
```

(E) $\begin{bmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{bmatrix}$

```
(%i25) determinant(E);
```

(%o25) $-b\left(b\left(b^2-a\,b\right)-a\left(b^2-a\,b\right)\right)-b\left(b\left(b^2-a\,b\right)-b\left(a\,b-b^2\right)+b\left(a^2-b^2\right)\right)+a\left(b\left(b^2-a\,b\right)-b\left(a\,b-b^2\right)+a\left(a^2-b^2\right)\right)+b\left(b\left(a\,b-b^2\right)-a\left(a\,b-b^2\right)\right)$

```
(%i26) factor(determinant(E));
```

(%o26) $-\left(b-a\right)^3\left(3\,b+a\right)$

3.6 (f)

(%i27)
```
F: matrix(
 [a+b,c,c],
 [b,c+a,b],
 [a,a,b+c]
);
```

(F) $$\begin{bmatrix} b+a & c & c \\ b & c+a & b \\ a & a & c+b \end{bmatrix}$$

(%i28)
```
determinant(F);
```

(%o28) $$(b+a)\left((c+a)(c+b)-a\,b\right)-c\left(b\,(c+b)-a\,b\right)+c\left(a\,b-a\,(c+a)\right)$$

(%i29)
```
factor(determinant(F));
```

(%o29) $4\,a\,b\,c$

4) oef 13

4.1 (a)

(%i30)
```
A: matrix(
 [2*x,2*y,x+y],
 [2*x,x+y,2*x],
 [x-y,2*y,2*x]
);
```

(A) $$\begin{bmatrix} 2\,x & 2\,y & y+x \\ 2\,x & y+x & 2\,x \\ x-y & 2\,y & 2\,x \end{bmatrix}$$

(%i31)
```
determinant(A);
```

(%o31) $$(y+x)\left(4\,x\,y-(x-y)(y+x)\right)+2\,x\left(2\,x\,(y+x)-4\,x\,y\right)-2\left(4\,x^2-2\,x\,(x-y)\right)y$$

(%i32)
```
factor(determinant(A));
```

(%o32) $$(y-x)^2\,(y+3\,x)$$

4.2 (b)

(%i33)
```
B: matrix(
 [m-1,6,m],
 [5,m,2],
 [-1,-3,1]
);
```

(B) $$\begin{bmatrix} m-1 & 6 & m \\ 5 & m & 2 \\ -1 & -3 & 1 \end{bmatrix}$$

(%i34)
```
determinant(B);
```

(%o34) $$\left(m-1\right)\left(m+6\right)+\left(m-15\right)m-42$$

(%i35)
```
factor(determinant(B));
```

(%o35) $$2\left(m-8\right)\left(m+3\right)$$

4.3 (c)

(%i36)
```
C: matrix(
 [p-4,p-1,p+2],
 [p-3,p,p+3],
 [p-2,p+1,p+4]
);
```

(C) $$\begin{bmatrix} p-4 & p-1 & p+2 \\ p-3 & p & p+3 \\ p-2 & p+1 & p+4 \end{bmatrix}$$

(%i37)
```
determinant(C);
```

(%o37) $$\left(p-4\right)\left(p\left(p+4\right)-\left(p+1\right)\left(p+3\right)\right)-\left(p-1\right)\left(\left(p-3\right)\left(p+4\right)-\left(p-2\right)\left(p+3\right)\right)+\left(p+2\right)\left(\left(p-3\right)\left(p+1\right)-\left(p-2\right)p\right)$$

(%i38)
```
factor(determinant(C));
```

(%o38) $$0$$

Opmerking: Dit hadden we op voorhand kunnen weten, want K2-K1 = [3 3 3]^T = K3-K2

4.4 (d)

(%i39)
```
D: matrix(
 [a,4,4],
 [4,a,a],
 [4,4,a^2]
);
```

(D) $$\begin{bmatrix} a & 4 & 4 \\ 4 & a & a \\ 4 & 4 & a^2 \end{bmatrix}$$

(%i40)
```
determinant(D);
```

(%o40) $a\left(a^3-4\,a\right)-4\left(4\,a^2-4\,a\right)+4\left(16-4\,a\right)$

(%i41)
```
factor(determinant(D));
```

(%o41) $\left(a-4\right)\left(a-2\right)\left(a+2\right)\left(a+4\right)$

4.5 (e)

(%i42)
```
E: matrix(
 [1,r,r^3],
 [1,s,s^3],
 [1,t,t^3]
);
```

(E) $$\begin{bmatrix} 1 & r & r^3 \\ 1 & s & s^3 \\ 1 & t & t^3 \end{bmatrix}$$

(%i43)
```
determinant(E);
```

(%o43) $-r\left(t^3-s^3\right)+s\,t^3+r^3\left(t-s\right)-s^3\,t$

```
(%i44) factor(determinant(E));
```

(%o44) $(s-r)(t-r)(t-s)(t+s+r)$

5) oef 14

```
(%i45) M: matrix(
         [m,m,m,m],
         [1,m,m,m],
         [1,1,m,m],
         [1,1,1,m]
       );
```

(M) $\begin{bmatrix} m & m & m & m \\ 1 & m & m & m \\ 1 & 1 & m & m \\ 1 & 1 & 1 & m \end{bmatrix}$

```
(%i46) factor(determinant(M));
```

(%o46) $(m-1)^3 m$

```
(%i47) solve(determinant(M)=m);
```

(%o47) $[m=-\frac{\sqrt{3}\,\%i-1}{2}, m=\frac{\sqrt{3}\,\%i+1}{2}, m=2, m=0]$

Antwoord: De reële oplossingen van deze vergelijking zijn m=0 en m=2.
(In deze cursus werken we niet met complexe getallen, dus de eerste twee oplossingen worden geschrapt.)

6) oef 15

```
(%i48) P: matrix(
         [1,p,p,p],
         [p,1,p,p],
         [p,p,1,p],
         [p,p,p,1]
       );
```

(P) $$\begin{bmatrix} 1 & p & p & p \\ p & 1 & p & p \\ p & p & 1 & p \\ p & p & p & 1 \end{bmatrix}$$

```
(%i49) factor(determinant(P));
```

(%o49) $-\left(p-1\right)^{3}\left(3\,p+1\right)$

```
(%i50) solve(determinant(P)=3*p+1);
```

(%o50) $[p=-\frac{\sqrt{3}\,\%i-3}{2},p=\frac{\sqrt{3}\,\%i+3}{2},p=-\frac{1}{3},p=0]$

Antwoord: De reële oplossingen van deze vergelijking zijn p=0 en p=-1/3.
(In deze cursus werken we niet met complexe getallen, dus de eerste twee oplossingen worden geschrapt.)